

# Chang He (Aaron)

aaron.he@student.unsw.edu.au | github.com/AaronHnoraA | aaronhnoraa.github.io | Sydney, Australia

## EDUCATION

---

### University of New South Wales (UNSW)

Sydney, Australia

*Bachelor of Science in Mathematics, UNSW 3956*

*Term 3, 2024 – Present*

- **Expected graduation:** Term 3, 2028 (Honours).
- **Planned transfer:** Bachelor of Computer Science, UNSW 3779.
- **Academic standing:** WAM **85**; admitted to the UNSW Talented Students Program (TSP) in 2025.

## RESEARCH INTERESTS

---

**Quantum computing:** quantum algorithms and quantum error correction, including fault-tolerant computation, stabilizer/CSS codes, and algebraic methods in finite-dimensional quantum systems.

**Mathematics:** algebraic methods for isomorphism problems, multilinear algebra and tensor spaces, group actions, canonical forms, and related representation-theoretic tools.

**Theoretical computer science:** computational complexity and algorithms, especially graph and tensor isomorphism, algebraic reductions, and structural problems at the interface of algebra and computation.

## RESEARCH EXPERIENCE

---

### Independent Study, UNSW

Sydney, Australia

*Quantum Computing, Algebraic Algorithms, and Isomorphism Problems with Youming Qiao* *Term 2, 2025 – Present*

- Conducting supervised reading and research preparation in quantum computation, complexity theory, and algebraic methods for isomorphism problems.
- Working on graph isomorphism and tensor isomorphism problems, with emphasis on algebraic formulations, reductions, and invariants for equivalence testing.
- Preparing a manuscript on connections between graph isomorphism and tensor isomorphism.

### Talented Students Program, UNSW

Sydney, Australia

*Academic Reading in Large Language Models with Quoc Le Gia*

*Term 1, 2025*

- Completed supervised academic reading through the UNSW Talented Students Program, focusing on selected literature in large language models.
- Developed practice in reading research papers, extracting technical assumptions, and presenting mathematical and algorithmic ideas clearly.